

Study Questions: Planning

1. What is the rostro-caudal gradient of temporal abstraction in the prefrontal cortex? What evidence supports it?
2. How does the basal ganglia implement policy selection through direct and indirect pathways?
3. What role does dopamine play in precision of policy selection?
4. How does Expected Free Energy decompose into pragmatic and epistemic value, and which PFC regions encode each?
5. What is frontal lobe syndrome? How does damage to PFC impair planning?
6. What is "utilization behavior," and how does Active Inference explain it?
7. How does Active Inference model impulsivity as low precision on long-horizon policies?
8. How does temporal discounting (preference for immediate rewards) relate to planning depth?
9. How does addiction progressively narrow the policy space in Active Inference terms?
10. How does depression lead to planning paralysis? What precision dynamics are involved?
11. Compare Active Inference's account of decision-making with classical expected utility theory.
12. What is the role of the anterior cingulate cortex (ACC) in effort-based decision-making and conflict monitoring?
13. How does working memory contribute to planning? What PFC regions support it?
14. What is the relationship between the default mode network (DMN) and planning/prospection?
15. How does theory of mind interact with planning in social contexts (strategic interaction)?
16. How might the prefrontal hierarchical gradient develop across childhood and adolescence?
17. How does Active Inference explain the "paradox of choice" — the finding that too many options can impair decision-making?

18. What is the computational difference between model-based planning (evaluating policies) and model-free habitual behavior?
19. How do orbitofrontal cortex (OFC) lesions affect decision-making? What Active Inference component is impaired?
20. Design a study using fMRI to test whether more abstract planning tasks activate more anterior PFC regions, as the temporal abstraction gradient predicts.